

VoiceRun: Voice-Powered Running App

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Abstract

- › About the project
- › Tasks & Storyboard
- › Prototype
- › Application
- › Conclusion



About the project •○

The problem



- While running, it's common to keep track of the **progress** using an application
- Many runners have found the interaction with such apps inconvenient, often requiring you to **physically touch** the screen
- To properly **check** the current information or to end the current session, **stopping** becomes almost mandatory



About the project ○●

The solution



- Using a **voice user interface** we can avoid the pitfalls of touch-based interface
- By using voice commands, users can keep their full attention on the activity **without interruptions**



Abstract

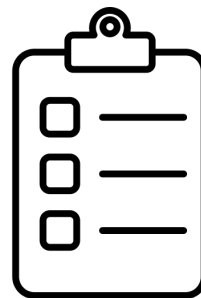
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Storyboard ●○○ Tasks



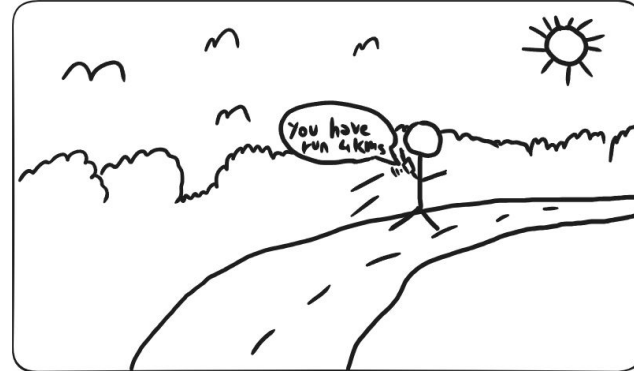
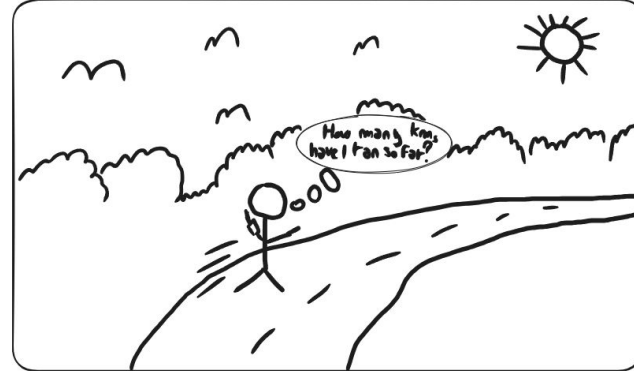
- The user wants to know:
 - their current **pace** mid-run
 - the current **distance** mid-run
 - the **time elapsed** since the beginning of the run
 - the **calories** burned during the run
- The user wants to view the **history** of their past runs
- The user wants to control the running session by **pausing**, **resuming**, or **stopping it**





Storyboard ○○○

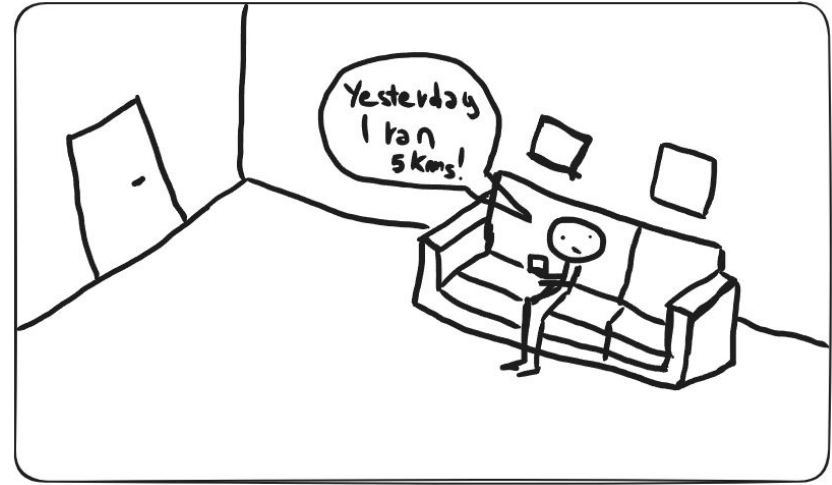
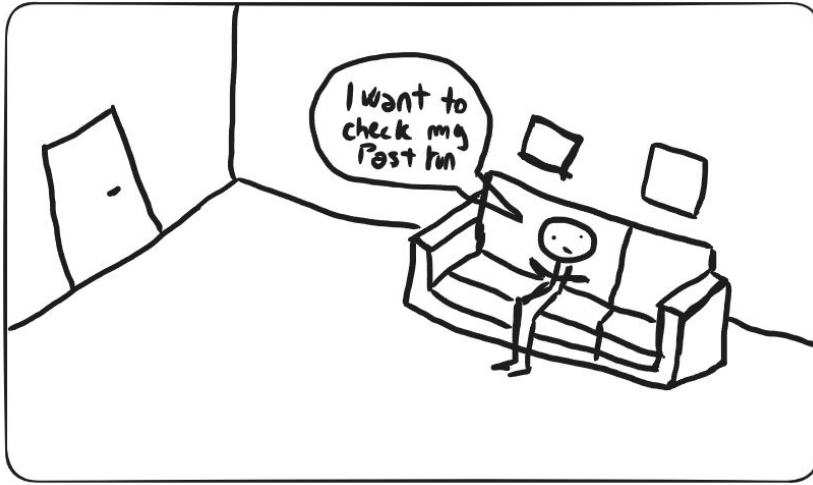
Storyboard - Vocal Interaction





Storyboard ○●●

Storyboard - Run History





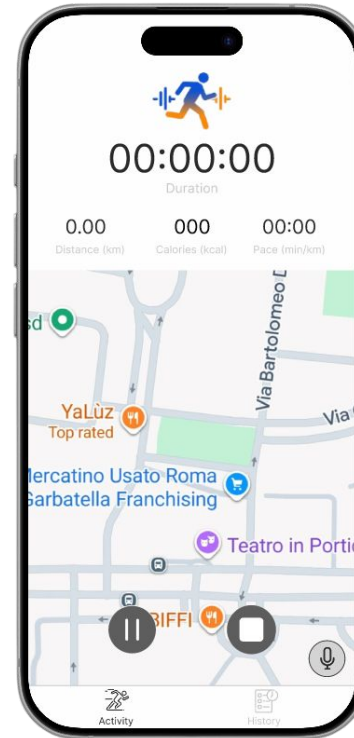
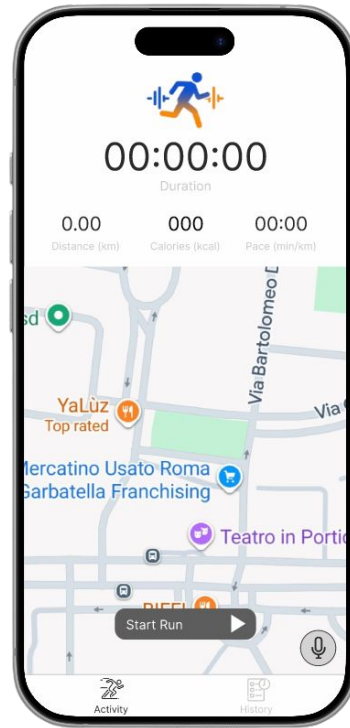
Abstract

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- › **Prototype**
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Prototype •○

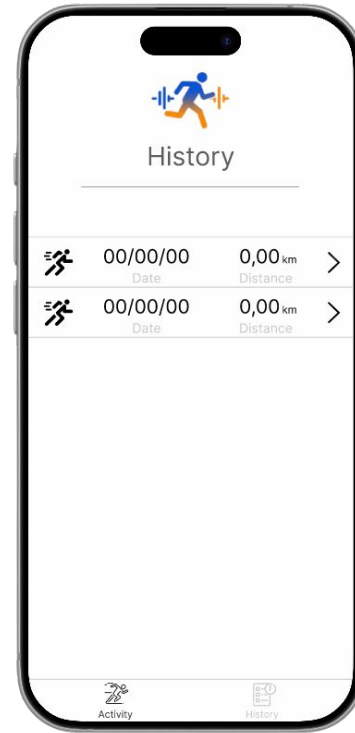
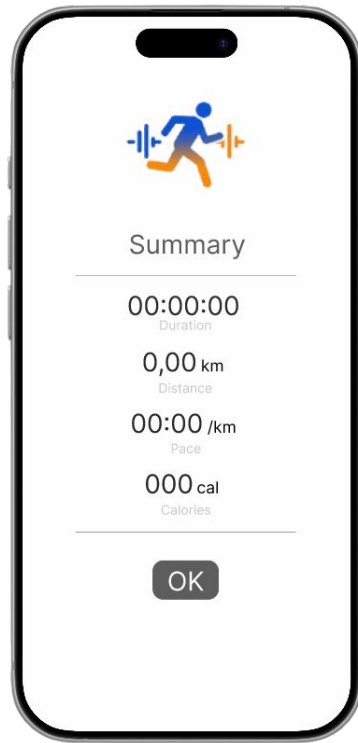
Prototype - Figma





Prototype ○●

Prototype - Figma





Abstract

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- › Prototype
- › **Application**
- › Conclusion

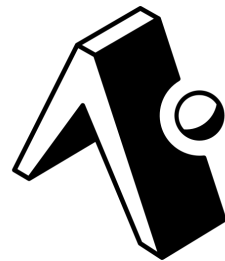
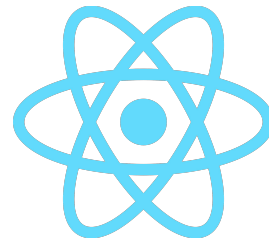


Application ●○○○○○○○○○○○

React Native & Expo



- The framework used to build our application is **React Native**
- We also used **Expo**, a **full-stack** framework to smoothen the development process of the app.
- Thanks to Expo it was possible to build the app for **both** Android and iOS.





Application ●○○○○○○○○○○○

Llama 3.1



- We used **Groq** for its fast inference time, while the actual language processing was handled by **Llama 3.1**

MODEL	REQUESTS PER MINUTE	REQUESTS PER DAY	TOKENS PER MINUTE	TOKENS PER DAY	ACTIONS
allam-2-7b	30	7K	6K	500K	
groq/compound	30	250	70K	No limit	
groq/compound-mini	30	250	70K	No limit	
<u>llama-3.1-8b-instant</u>	30	14.4K	6K	500K	
llama-3.3-70b-versatile	30	1K	12K	100K	
meta-llama/llama-4-scout-17b-16e-instruct	30	1K	30K	500K	
meta-llama/llama-prompt-guard-2-22m	30	14.4K	15K	500K	
meta-llama/llama-prompt-guard-2-86m	30	14.4K	15K	500K	
openai/gpt-oss-120b	30	1K	8K	200K	

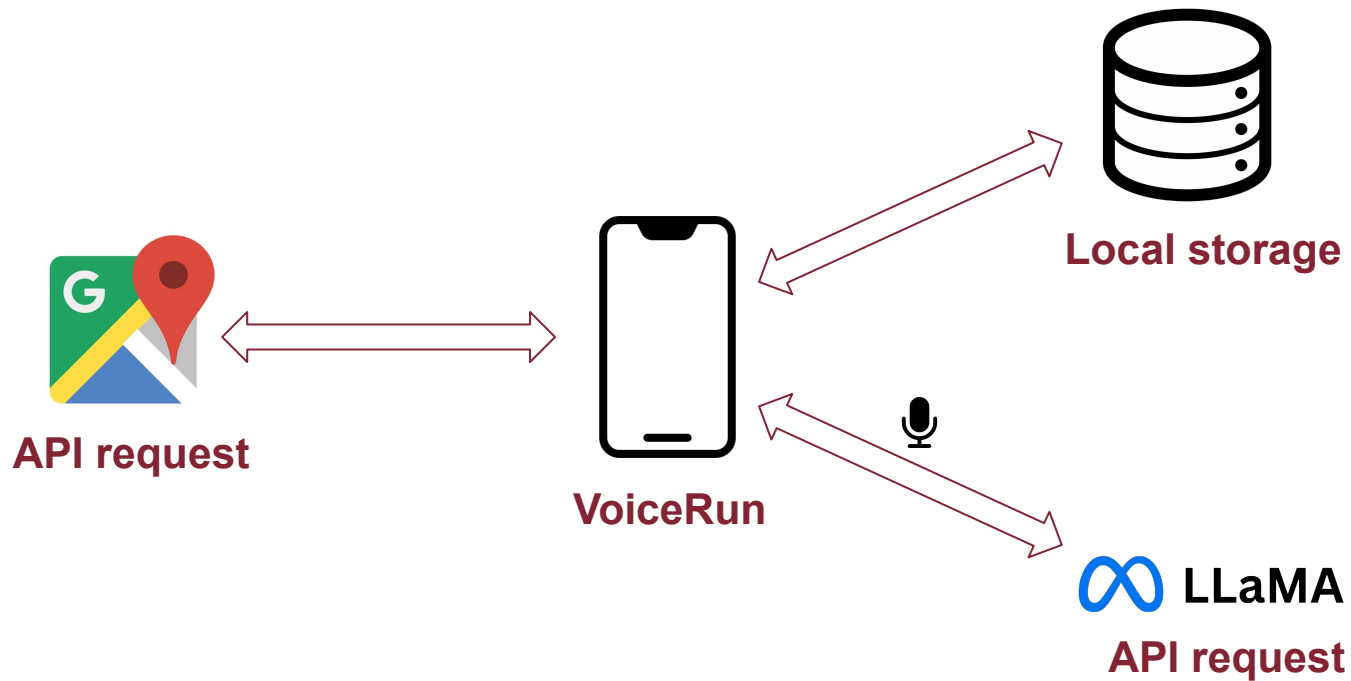
 LLaMA





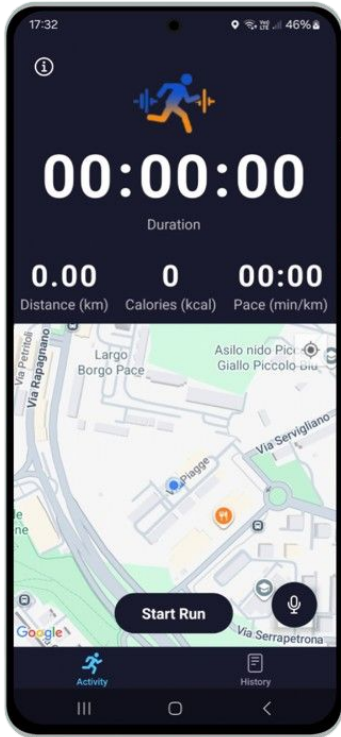
Application ○●○○○○○○○○○

App Structure





Application ○○○●○○○○○○○○○
Activity Tab



Start an activity, then **stop/resume** or **end** it

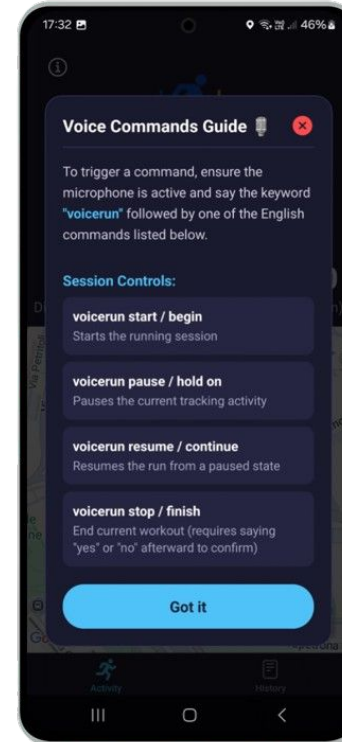
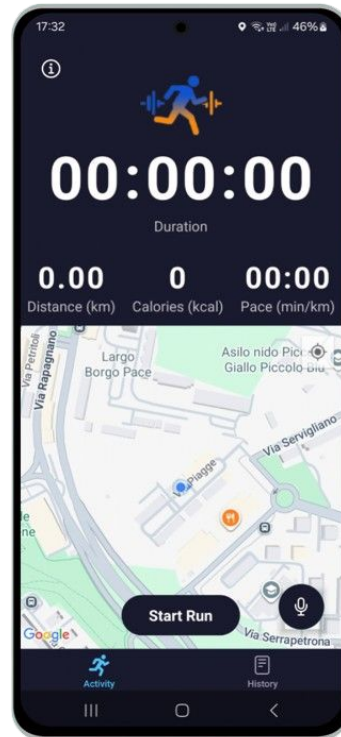
Ending an activity shows its **summary**

You can use vocal commands to make the app talk about requested information (**pace, km, time**, ecc...)



Application ○○○○●○○○○○○

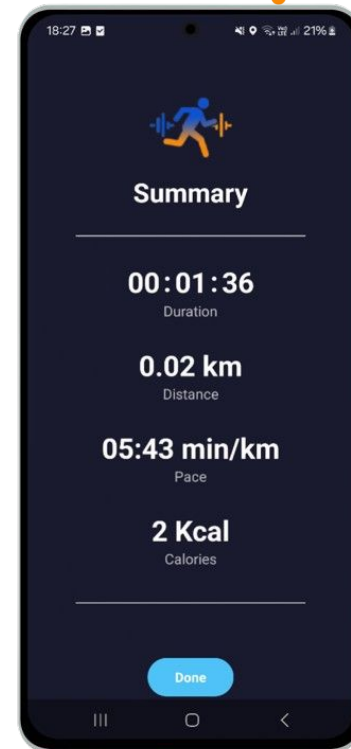
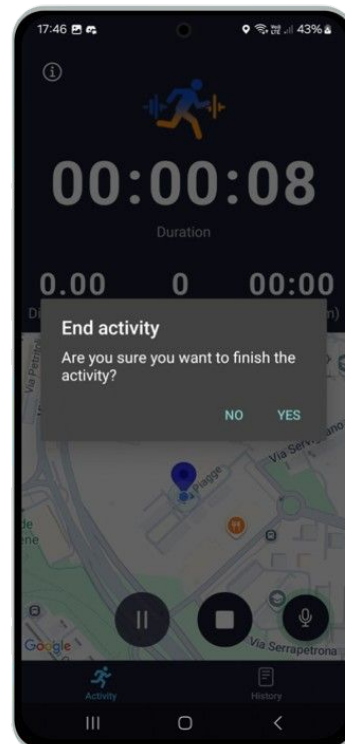
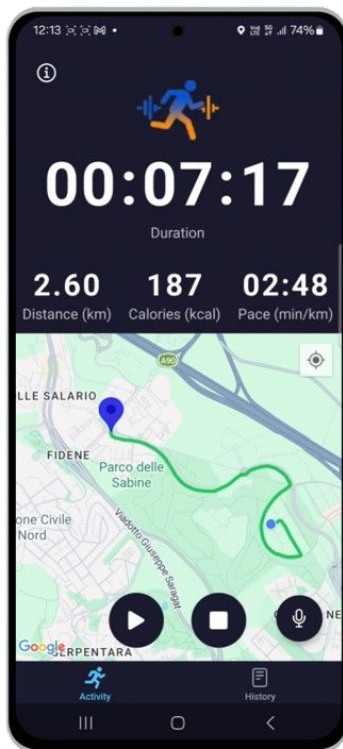
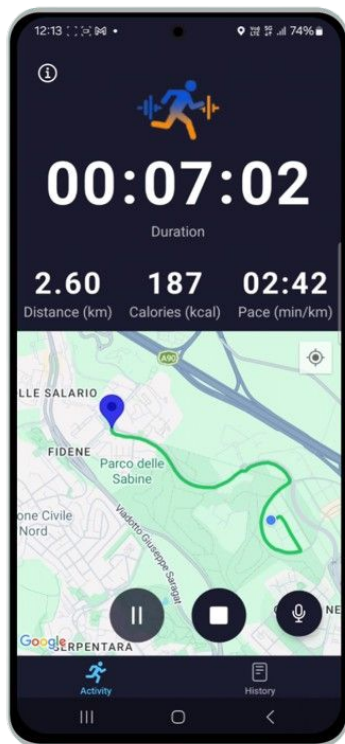
Activity Tab - Commands Guide





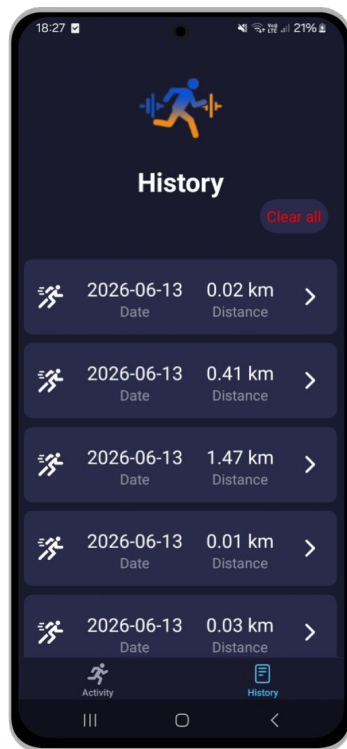
Application ○○○○○●○○○○○

Activity Tab - While Running





Application ○○○○○●○○○○○ History Tab



Check past runs

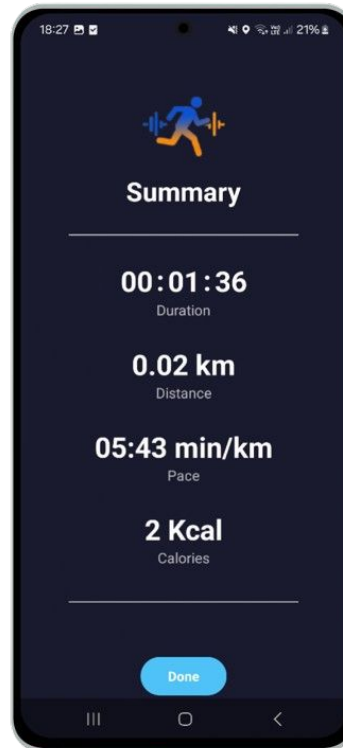
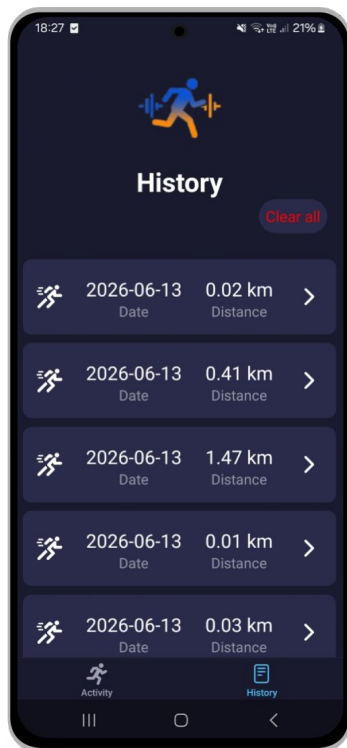
Click on a run to access its **summary**

Delete all runs



Application ○○○○○○●○○○

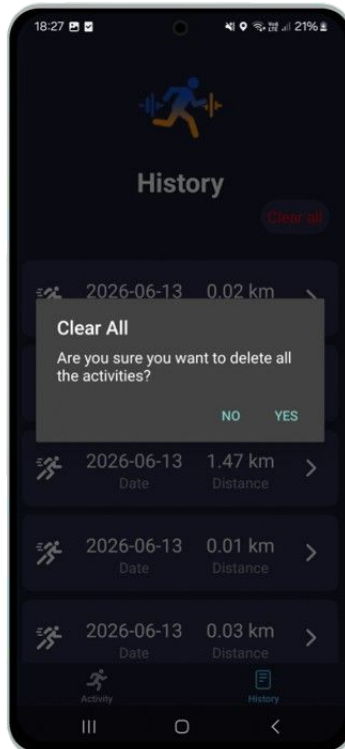
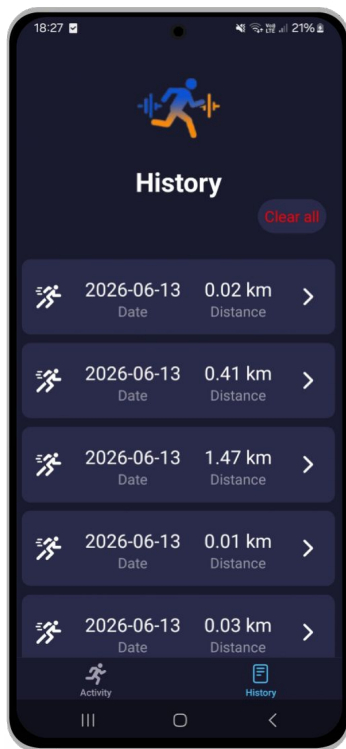
History Tab - Summary





Application ○○○○○○○●○○○

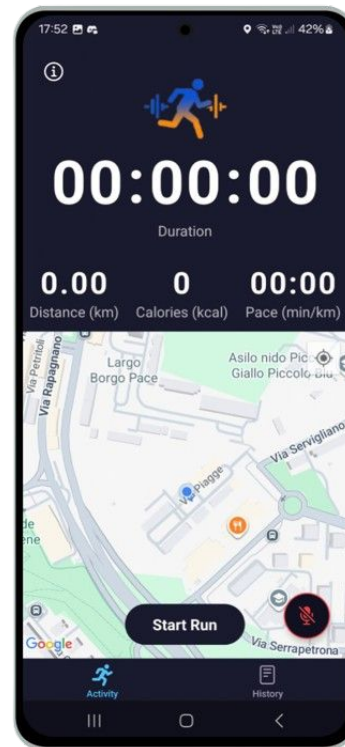
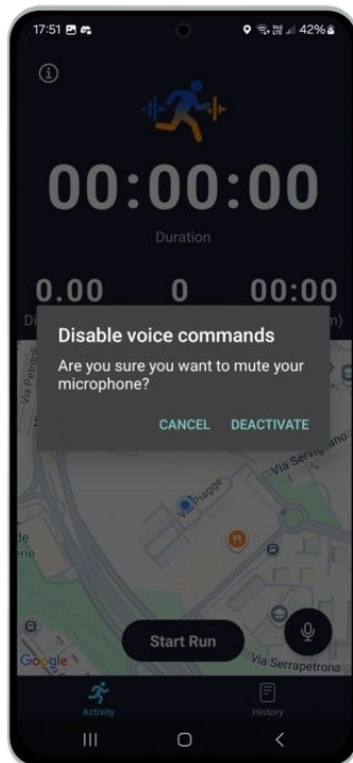
History Tab - Clear All





Application ○○○○○○○○●○○○

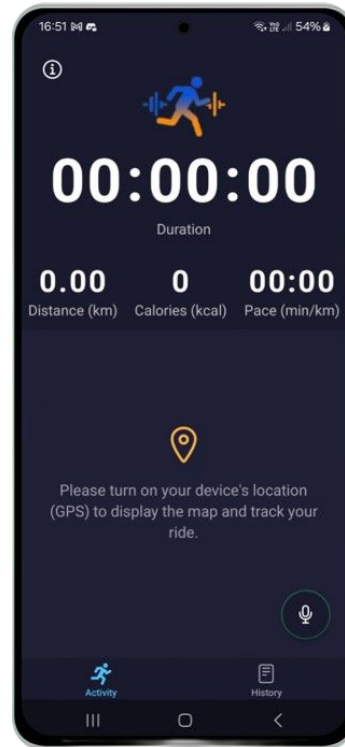
Edge Cases - Disable Microphone





Application ○○○○○○○○○●○

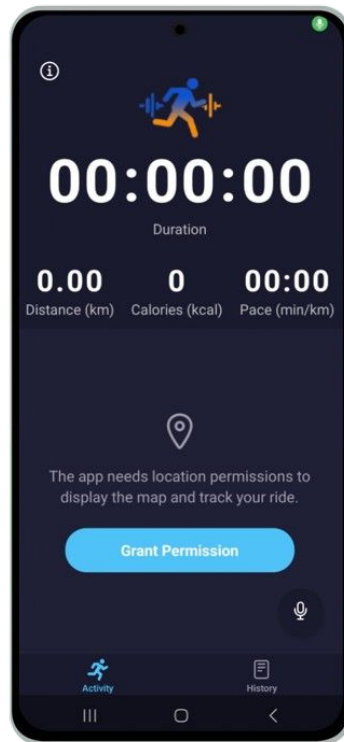
Edge Cases - Turn Off Location





Application ○○○○○○○○○○●

Edge Cases - Location Permissions





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Conclusion •

Thoughts and future developments



- VoiceRun performs its tasks **well**, allowing users to use their voice for interactions
- For possible future developments, **increasing the receptivity** and understanding of the vocal commands could lead to a **smoother user experience**
- Beyond that, using other paid models could allow the app to actually be distributed without having to worry about **tokens limit**



VoiceRun: Voice-Powered Running App

Thank you for the attention!